UNIVERSITÄT BERN

Assignment

Derivatives

1. Common Part

a)

To answer the first question, the Black–Scholes Model is applied and the numbers are filled. The Black–Scholes call option price is:

$$C = 356.72$$

b)

First, the simulations are done and the differences to the price calculated in question a) can be computed:

Simulations	Price	Difference to a)
100	\$483.14	483.14 – 356.72 = \$126.42
1000	\$393.48	393.48 – 356.72 = \$36.76
100,000	\$358.32	358.32 – 356.72 = \$1.60
500,000	\$356.57	356.57 – 356.72 = -\$0.15

Tabelle 1.1: Monte Carlo option prices vs. Black–Scholes benchmark.

The difference gets smaller the more simulations are made. This is due to the statistical nature of the Monte Carlo method, which estimates the value of an option using simulated price paths. The more simulations are done, the more paths are included in the estimation. Thus, the statistical error or standard deviation decreases with more simulations.

c)

The price at maturity should be equal or more than \$7,392.96 for the option to outperform the long position in the underlying. This is computed by setting the net return of the option equal to or larger than the return of the underlying. The return achieved at \$7,392.96 is 10.16%.

d)

To answer this, the price of the call option is again calculated using the Black–Scholes model but with a continuously compounded interest rate of 5%. This yields a call price of:

$$C = 448.07$$

Using the same method as in question c), the price at maturity needed for an outperformance is:

$$S_T = 7{,}500.81, \qquad \text{return} = 11.77\%$$

The price is higher than with the 2% rate, because the interest rate is used to discount the option to its present value. A lower rate results in a smaller discount and therefore a smaller call price.

e)

To implement a costless insurance strategy, we go long in the index, obtain a 1-year put with strike $0.9S_0$, and we short a 1-year call with a strike such that both premiums are offset. This caps the downside loss at 10%. The maximum upside lies at the call strike. Thus, the maximum payoff of the total position after one year is $7'851,98.

f)

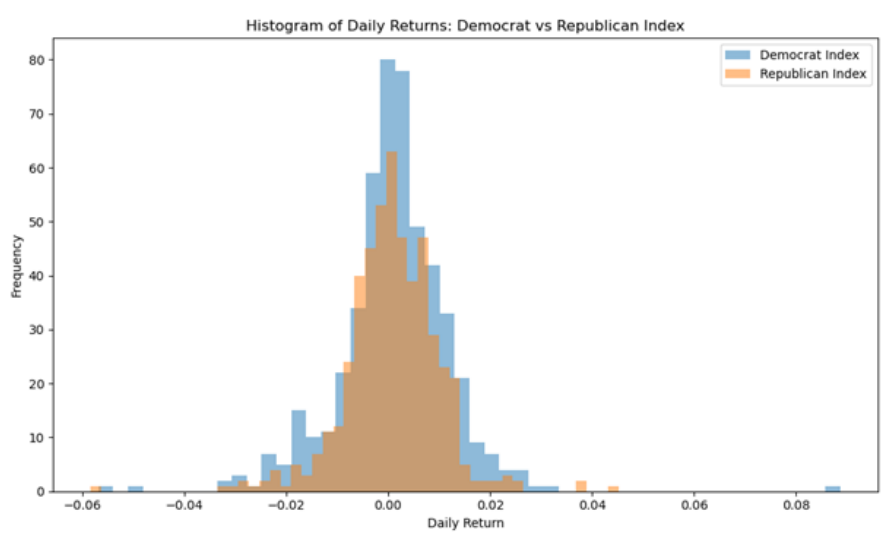
The price of the down-and-in barrier product is:

$$\text{Down-and-In Call Price} = 414.67913361123647$$

The cost of the insurance strategy in question (e) is offset by giving up possible gains above the cap of the call option. The downside protection is always in place, no matter how much the underlying drops. The down-and-in barrier version only gets "activated" if the price of the underlying drops below $6,400, resulting in a lower price than the strategy in question (e). This is, in fact, the same idea as the pay-later strategy discussed in class.

g)

The histogram of the daily returns of the time series looks as follows.



We cannot objectively say which is better. A risk-averse investor would choose the Republican Index due to lower volatility (0.009 vs. 0.011). A return-maximizing investor

Statistic	Democrat Index	Republican Index
Mean	0.0011	0.0008
Volatility (Std)	0.0112	0.0093
Skewness	0.2405	-0.4237
Kurtosis	9.2956	5.3753
5th Percentile	-0.0183	-0.0132
95th Percentile	0.0161	0.0132

Tabelle 1.2: Summary Statistics for Democrat and Republican Index Returns

would choose the Democrat Index due to the higher mean (0.001 vs. 0.0008), accepting higher risk.

h)

We calculated the price of two puts using 100,000 simulations. The payoff used was:

$$\text{payoff} = \max(K - S_{\text{evolution}}[0, i],\ 0)$$

Results:

$$\text{Put Democrat Index} = 1.099574822889297$$

$$\text{Put Republican Index} = 0.6381167466857145$$

They differ because the underlyings have different volatilities and price levels.

i)

The American put was computed with 100,000 simulations. The payoff was updated to:

$$\text{payoff} = \max(\text{payoff},\ K - S_{\text{evolution}}[0, i])$$

Result:

$$\text{American Put Republican Index} = 1.2000221636489685$$

Maximum tariff for the European option (difference to European put):

$$\text{Tariff} = 0.561905416963254$$

j)

For the lookback option simulation, the payoff was modified as follows:

$$S_{\min} = \min S_{\text{evolution}}, \qquad \text{payoff} = \max(K - S_{\min}, 0)$$

Result:

$$\text{Lookback Put Democrat Index} = 2.0482170830097743$$

The lookback option is more expensive because its payoff depends on the minimum price during the option's life, increasing expected payoff. It is more sensitive to volatility, as higher volatility increases the chance of a very low minimum.

2. Individual Part

a)

The underlying of the product is Alphabet, Inc. The spot reference price at issuance is USD 195.28, and the strike price is also set at USD 195.28, corresponding to 100% of the spot. The product references 5.12085 underlyings in total.

This product is an exotic European call payoff embedded in a capital-protected structured note. The derivative component behaves like a European call because the payoff depends only on the final fixing and increases linearly above the strike price, but it is considered exotic rather than vanilla since the exposure is scaled: the note references 5.12085 underlyings, which corresponds to 2.407 effective calls once the 47% participation rate is applied. The payoff therefore has a non-standard notional and partial participation, which makes it non-vanilla even though it has no barriers and no path dependence. Economically, the product can be replicated as a zero-coupon bond with a maturity value of 970 together with 2.407 classical European long calls struck at USD 195.28.

The product pays off at maturity, when the final fixing of the underlying is observed. It has no upside capreturns can increase without limit if the underlying finishes above the strike. On the downside, it includes a floor through its 97% capital protection, ensuring a minimum redemption of USD 970 even if the underlying performs poorly.

The position is a long call.

b)

In this part, we price the Vontobel Capital Protection Note with Participation on Alphabet Inc. using a Monte-Carlo simulation approach. The payoff at maturity is specified in the Final Terms as

$$\text{Redemption} = \text{Capital Protection} + \max\{(SF - X) * B * P,\, 0\},$$

where X is the strike, SF the final fixing price, B the number of underlyings, and P the participation rate. At issuance, the issuer states that the net present value (NPV) of the bond component (The capital protection) equals

$$\text{NPV}_{\text{bond}} = 950.198.$$

Thus, the product can be decomposed into a bond and an embedded call-like component. Our task is to price the latter using simulation.

Model Parameters

We use the parameters provided in the Factsheet and the assumptions specified below:

$$\begin{aligned} S &= 195.28, & K &= 195.28, \\ B &= 5.12085, & P &= 0.47, \\ T &= 0.5, & \sigma &= 0.3324, \\ \delta &= 0.003, & rf &= 0.0412. \end{aligned}$$

The value $B \cdot P = 5.12085 \times 0.47 \approx 2.4$ indicates that the payoff corresponds to the payoff of roughly 2.4 at-the-money European call options on Alphabet, in addition to a capital-protected bond component. This interpretation also provides a useful plausibility check for the simulated option value.

Market Assumptions

Risk-Free Rate: We use the *6-month U.S. Treasury Bill rate* (U.S. Department of the Treasury 2025) at the initial fixing date, $r_f = 4.12\%$. Since the product has a maturity close to six months, this rate is the correct benchmark for discounting under the risk-neutral measure.

Dividend Yield: According to Yahoo-Finance, Alphabet does pay dividends of $\delta = 0.3\%$. Thus we set $\delta = 0.3\%$. (Yahoo Finance 2025)

Volatility: We estimate the volatility using the Yahoo-Finance API to find the *historical six-month volatility* of Alphabet from 6.2.2025 to 6.8.2025, yielding $\sigma = 0.3324$. Given that the derivative itself has a six-month maturity, matching the estimation window to the horizon is reasonable. You could also calculate the implied volatility on the initial fixing date using a call option and the Black and Scholes- Model. However the implied volatility may incorporate option-market expectations, but might also be distorted by short-term supply–demand effects. Furthermore the implied volatility would be very close to the historical volatility anyway, so it wouldn't make much of a difference to use it here.

Simulation Framework

We follow the risk-neutral geometric Brownian motion (GBM) dynamics:

$$S_{t+h} = S_t \exp\left((r_f - \delta - \tfrac{1}{2}\sigma^2)h + \sigma\sqrt{h}\,\varepsilon\right), \qquad \varepsilon \sim N(0,1),$$

with daily discretization as required:

$$\text{trading days per year} = 250, \qquad N = T \cdot 250, \qquad h = \frac{T}{N}.$$

We run 100000 simulations to estimate the expected discounted payoff

$$\mathrm{NPV}_{\text{option}} = e^{-r_f T}\,\mathbb{E}\left[\max\{(SF - K) * B * P,\, 0\}\right],$$

and obtain the total price as

$$\text{Price} = \mathrm{NPV}_{\text{bond}} + \mathrm{NPV}_{\text{option}}.$$

With these assumptions we get a Price of 998.5148

c)

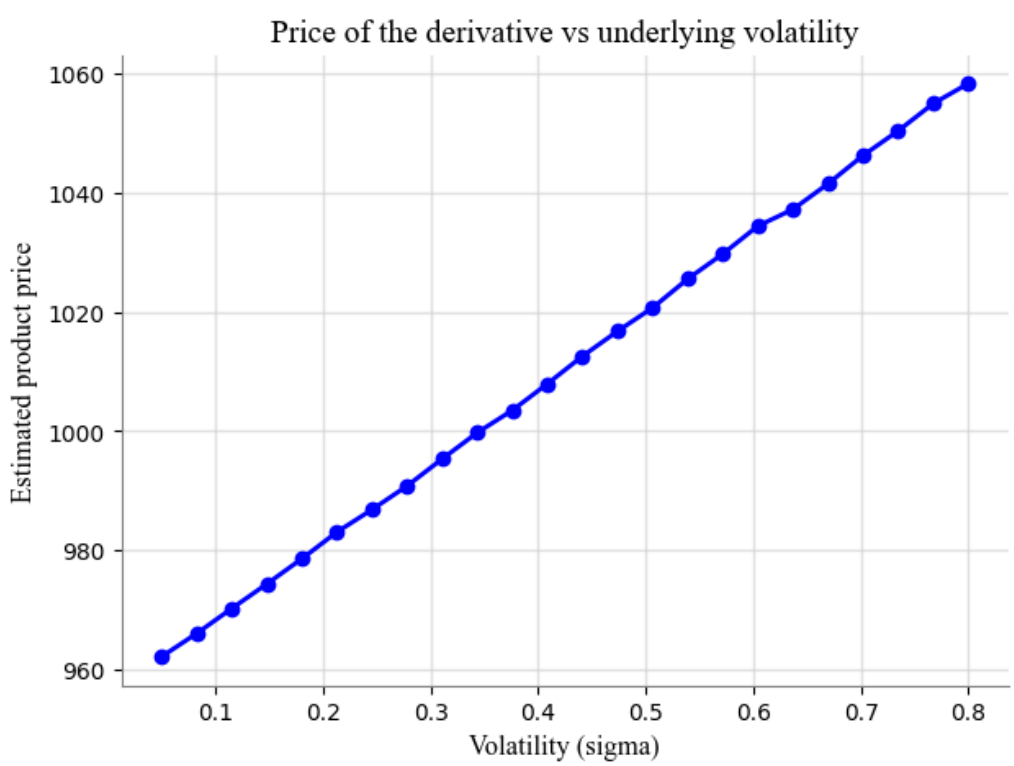
The structured product can be decomposed into two components:

$$\text{Product Price} = \text{NPV of bond} + \text{value of a call-like payoff.}$$

The bond component is fixed and independent of volatility. Therefore, the only volatility–sensitive part is the option component

$$\max\{(S_T - K) * B * P, 0\},$$

which is strictly increasing in the volatility of the underlying asset. A higher volatility increases the probability of large upward movements in the underlying price while the downside is limited by the capital protection. Thus, from the perspective of an investor who holds the product to maturity, a higher volatility is beneficial.
This effect is confirmed by the Monte Carlo results. The figure below plots the estimated price of the derivative for different volatility levels while keeping all other inputs constant.



The plot shows a clear, approximately linear upward-sloping relationship between volatility and the simulated price of the derivative. The price increases from roughly \$960 at low volatility levels ($\sigma = 0.05$) to more than \$1,050 when volatility rises to $\sigma = 0.80$.

Conclusion Since the investor is effectively long a call option embedded in a capital-protected note, higher volatility increases the expected upside without affecting the guaranteed minimum payoff. Therefore, an investor holding this product would *prefer a higher volatility of the underlying* during the investment period.

d)

Our simulation yields a theoretical fair value of

$$\hat{V}_{\text{model}} = 998.5148,$$

while the issuer's price is

$$V_{\text{market}} = 1000.$$

The difference of approximately USD 1.50 USD is negligible and lies well within normal model uncertainty such as volatility estimation errors, discretization effects, and numerical variation from the Monte-Carlo simulation. Therefore, *this deviation does not constitute an arbitrage opportunity.* In a frictionless market, arbitrage would require a strictly lower replication cost that can be traded with certainty, which is not the case here.

Pros

Capital protection. At maturity, 97% of the notional is guaranteed, which substantially limits downside risk.

Upside participation. The embedded option component corresponds to roughly 2.4 at-the-money call options on Alphabet, enabling the investor to benefit from positive price movements.

Fair pricing. Our model price is almost identical to Vontobel's issue price, indicating that the product is neither materially overpriced nor underpriced.

Potential benefit from rising volatility. If implied volatility increases after purchase, the value of the embedded call options increases as well.

Cons

Limited participation. Only 47% of the positive performance of the underlying is passed on to the investor; strong rallies are only partially captured.

Low payoff in sideways markets. If Alphabet remains near the strike, the participation component contributes little to the overall payoff.

Market Expectations Looking ahead, we consider a rise in volatility to be plausible due to the ongoing "AI boom" and the increasing competitive pressure in the technology and cloud-computing sectors. Alphabet is heavily involved in AI development, and uncertainty surrounding technological breakthroughs tends to increase market volatility. Higher volatility would raise the value of the embedded call options, making the product more valuable after purchase. While we avoid hindsight bias in pricing, this forward-looking expectation leads us to view the product favorably.

Recommendation: Buy. The Product is fairly priced, offers substantial downside protection, and stands to benefit if volatility increases, which we consider a realistic scenario given current market dynamics in the AI sector. Therefore, based on both valuation and market expectations, we recommend purchasing the option at the calculated price.

Bibliography

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[2] Yahoo Finance (2025), Financial market data and market overview. Available online at: https://finance.yahoo.com/, Accessed on November 17, 2025

statement of authorship

We hereby declare that each of us has contributed their part to the solution of this assignment without any help from others. Bern, 1. Dezember 2025

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